

Topic 3

Functions

**Concept guide · Worked examples · Practice · Investigation and modelling
· Solutions**

Learning focus

Content explanations, original Cambridge-style practice and an investigation task tailored to International Mathematics 0607.

Topic map

3.1	Graphs of functions and representations
3.2	Function notation and domain
3.3	Composite and inverse functions
3.4	Linear, quadratic and reciprocal graphs
3.5	Transformations of function graphs
3.6	Graphical solutions and modelling

Core foundations and Extended applications are integrated. Practise exact methods, calculator work and the written reasoning required in investigations.

1. Functions as input-output rules

A function assigns one output to each allowed input. Its domain gives permitted inputs; its range gives outputs. In $f(x)$, replace every x with the given value before evaluating. A table, equation, graph and verbal rule can describe the same relationship.

Worked example

If $f(x) = x^2 - 3x$, then $f(-2) = (-2)^2 - 3(-2) = 10$.

Exam check: A single input cannot have two different outputs in a function.

2. Domain, intercepts and graph shapes

Linear graphs are straight lines, quadratics are parabolas, and reciprocals have excluded values and asymptotes. Set $x=0$ to find a y -intercept and set $f(x)=0$ for x -intercepts. Look at the domain before drawing a curve: $y=1/x$ is undefined at $x=0$.

Worked example

For $f(x)=x^2-4$, intercepts are $(-2,0)$, $(2,0)$, and $(0,-4)$.

Exam check: Plotting points alone does not justify a straight line when a curve is needed.

3. Compositions and inverses

The composition $fg(x)=f(g(x))$ applies g first, then f . A one-to-one function has an inverse f^{-1} that reverses its mapping; to find it, set $y=f(x)$, interchange x and y and solve for y . State domain restrictions when needed for a quadratic inverse.

Worked example

$f(x)=2x+3$ and $g(x)=x^2$: $f(g(2))=f(4)=11$, while $g(f(2))=g(7)=49$.

Exam check: Order matters: $fg(x)$ and $gf(x)$ are usually different.

4. Graph transformations and intersections

For $y=f(x)+k$ shift a graph vertically by k ; $y=f(x-a)$ moves it right by a . Graphical solutions to $f(x)=g(x)$ occur at intersections. With a graphic display calculator, choose a window showing important intercepts and turning points, then verify approximate values by substitution.

Worked example

For $f(x)=x^2$, $y=f(x-3)+2$ is $y=(x-3)^2+2$, with vertex $(3,2)$.

Exam check: A calculator plot outside a suitable window can hide roots or intersections.

5. Quadratic and exponential models

A quadratic model changes direction at a turning point; an exponential model multiplies by a constant factor per equal step. Choose a model from data by testing first or second differences or successive ratios, then compare predictions with the original observations. State when extrapolation may fail.

Worked example

Values 5,10,20,40 at times 0,1,2,3 fit $f(t)=5 \times 2^t$. At $t=4$, predict 80.

Exam check: Four agreeing observations suggest a model but cannot prove it for every future time.

6. Graphical methods

Use a plot to estimate an intersection or turning point, but show the function equations and read a scale. If software gives $x \approx 1.73$, round as required and check both sides of an equation near that x . Distinguish an exact root found algebraically from a numerical estimate.

Worked example

For $x^2=9$, the intersections of $y=x^2$ and $y=9$ are at $x=-3$ and $x=3$.

Exam check: Inspect both sides of a quadratic graph so that negative roots are not missed.

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Graphical calculator and Paper 6

Select a graph window that shows intercepts and turning points. Verify numerical intersections by substitution, and state domain and units.

State the rule or model, display enough working to reproduce your result, test it against the given values and discuss at least one limitation.



Practice paper

Topic 3 · Functions

Instructions

Original practice, informed by June 2026 components. Section A: short multiple choice for revision without a calculator. Section B: show full written working. Section C: investigation and modelling with reasons and checks.

Candidate name:	
Class / date:	

Total: 40 marks

Section A · Multiple choice

Choose one answer for each question.

1. For $f(x)=2x+1$, find $f(3)$. [1]

- A 5
- B 6
- C 7
- D 8

2. For $f(x)=x^2$, find $f(-4)$. [1]

- A -16
- B -8
- C 8
- D 16

3. Which x is excluded from $f(x)=1/(x-5)$? [1]

- A -5
- B 0
- C 1
- D 5

4. If $f(x)=3x$ and $g(x)=x+2$, find $fg(1)$. [1]

- A 5
- B 7
- C 9
- D 11

5. The inverse of $f(x)=2x-1$ is [1]

- A $(x+1)/2$
- B $2x+1$
- C $(x-1)/2$
- D $1/(2x-1)$

6. The graph $y=f(x)+4$ shifts [1]

- A left 4
- B right 4
- C up 4
- D down 4

7. What shape is $y=x^2-1$? [1]

- A straight line
- B parabola
- C circle
- D reciprocal hyperbola

8. Where do graphical solutions of $f(x)=g(x)$ occur? [1]

- A y-axis intercepts
- B turning points only
- C intersections
- D asymptotes

Section B · Structured questions

9. The function $f(x)=x^2-4x+3$ is defined for all real x .

(a) Find $f(0)$. [1]

(b) Find the roots of $f(x)=0$. [2]

(c) Find the vertex by completing the square. [3]

[Total: 6]

10. Functions $f(x)=2x+5$ and $g(x)=x^2$ are given.

(a) Find $f(g(3))$. [2]

(b) Find $g(f(3))$. [2]

(c) Find $f^{-1}(x)$.

[2]

[Total: 6]

11. A water tank starts with 100 L and loses 15 L each minute at a constant rate.

(a) Write $V(t)$ for its volume after t minutes.

[2]

(b) Find when $V(t)=40$.

[2]

(c) State a sensible domain for the model.

[2]

[Total: 6]

12. The functions $p(x)=x^2$ and $q(x)=2x+3$ intersect.

(a) Write an equation for their intersections.

[1]

(b) Find both x-coordinates.

[3]

(c) Find the corresponding y-coordinates.

[2]

[Total: 6]

Section C · Investigation and modelling

Show the rule, test it with values and explain the result in context.

13. A population model gives $P(t)=500 \times 1.2^t$, with t in years.

(a) Find $P(0)$ and $P(1)$. [2]

(b) Explain what 1.2 means in context. [2]

(c) Estimate $P(3)$ to the nearest whole number. [2]

(d) Explain why extrapolating this model far into the future may be unreliable. [2]

[Total: 8]



Worked solutions

Fold back this section while completing the practice paper.

Marking routine

Method, model, communication and verification matter. A numerical answer should follow from clear working.

Section A · Answers and reasons

1. C — 7. Substitute $x=3$: $2(3)+1=7$.
2. D — 16. The square of -4 is positive 16.
3. D — 5. $x=5$ makes the denominator zero.
4. C — 9. Apply g first: $g(1)=3$, then $f(3)=9$.
5. A — $(x+1)/2$. Reverse $y=2x-1$: $x=(y+1)/2$.
6. C — up 4. Adding 4 to each output moves the curve up.
7. B — parabola. The graph of a quadratic is a parabola.
8. C — intersections. Common x,y pairs occur where graphs cross.

Section B • Worked solutions

9. a) 3 [1]. b) $x^2-4x+3=(x-1)(x-3)$, so $x=1,3$ [2]. c) $(x-2)^2-1$ gives vertex $(2,-1)$ [3].

10. a) $g(3)=9$, $f(9)=23$ [2]. b) $f(3)=11$, $g(11)=121$ [2]. c) $y=2x+5 \Rightarrow x=(y-5)/2$, so $f^{-1}(x)=(x-5)/2$ [2].

11. a) $V(t)=100-15t$ [2]. b) $100-15t=40 \Rightarrow t=4$ min [2]. c) Volume must be nonnegative: $0 \leq t \leq 20/3$ min [2].

12. a) $x^2=2x+3$ [1]. b) $x^2-2x-3=(x-3)(x+1)=0$, so $x=3,-1$ [3]. c) $y=9$ and $y=1$ respectively [2].

Section C • Investigation and modelling solutions

13(a). $P(0)=500$, $P(1)=600$ [2].

13(b). The model assumes growth by 20% each year [2].

13(c). $500(1.2)^3=864$ [2].

13(d). Resources, capacity or growth rate may change; the constant multiplier is an assumption [2].

Self-check

Explain why your model fits the starting data, show an independent check and name a condition under which your prediction would be unreliable.

Reference: Cambridge IGCSE International Mathematics 0607 syllabus 2025-2027, <https://www.cambridgeinternational.org/Images/662472-2025-2027-syllabus.pdf>. Assessment conventions informed by the June 2026 papers and mark schemes supplied by the user. All practice questions are newly written.